EPICODE

Final Project Assignment

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Project expectations: goal

Develop a well-architected, secure, and maintainable Java SE application showcasing object-oriented design, use of patterns, and core Java technologies.

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Project expectations: minimum requirements

4 Required Design Patterns

Factory, Composite, Iterator, Exception Shielding

Core Technologies

Collections, Generics, Java I/O, Logging, JUnit Testing

Secure Programming

Input Sanitization, No Hardcoded Secrets, Controlled Exception Propagation

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Project expectations: aiming for the maximum

Beyond Minimum

Additional patterns and technologies from a defined list.
Correct integration and critical-path usage of optional components.
Additional points do not raise the grade beyond 50, but allow varied strategy.

Security Penalties

Crashes, stack traces, insecure data handling will subtract points.

Documentation and Justification

README must include project overview, design decisions, diagrams, tech/pattern justifications, limitations.
Expect interview questions on alternatives and rationale.

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Project expectations: Freedom of Design

Students are free to select any project scenario or application domain of their choice. This could be a utility, a simulation, a personal productivity tool, a data processing tool, or anything else that fits the criteria.

The minimum required features must be implemented as specified.

Students may integrate any number of advanced technologies or design patterns from the optional list to improve their score.

Projects will be evaluated not just on what is included, but how well technologies are integrated, justified, and secured.

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Project expectations: deliverables

1. Complete Java project source code
2. README.md with:
   a. Application overview and functionality
   b. Technologies and patterns used, with justification
   c. Setup and execution instructions
   d. UML diagrams (class + architectural)
   e. Known limitations and future work
3. Test suite

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Project expectations: interview

During the evaluation, you will:

1. Demo your application live
2. Walk through random parts of the code
3. Justify your design decisions

Expect questions on why specific technologies and patterns were used over alternatives. Your understanding and rationale are part of your grade.

Evaluation Breakdown

Total: α

Constraints
$25 \leq \alpha \leq 30$

Required Features

Design Patterns	**(16 pts total)**
Factory	(3 pts)
Composite	(4 pts)
Iterator	(4 pts)
Exception Shielding	(5 pts)
Technologies	**(14 pts total)**
Collections Framework	(3 pts)
Generics	(3 pts)
Java I/O	(3 pts)
Logging	(2 pts)
JUnit Testing	(3 pts)

All these items are required:
(you must include all of them in your project)

Each item coding and usage will be evaluated from 0 to N points.

You must obtain a minimum of 25 points from these items.
(not everything must be perfect)

Evaluation Breakdown

Total: $\alpha + \beta$

Constraints
$25 \leq \alpha \leq 30$
$0 \leq \beta \leq 25$

Optional Advanced Features

Technologies

Multithreading	(8 pts)
Stream API & Lambdas	(5 pts)
Reflection	(4 pts)
Inversion of Control	(5 pts)
Custom Annotations	(2 pts)
Mockito	(3 pts)

Design Patterns

Abstract Factory	(3 pts)
Builder	(3 pts)
Strategy	(4 pts)
Observer	(5 pts)
Chain of Responsibility	(4 pts)
Adapter/Bridge/Proxy/Decorator	(2 pts each)
Singleton	(2 pts)
Memento	(3 pts)
Template Method	(3 pts)

Evaluation Breakdown

Total: $\alpha + \beta - \gamma$

Constraints

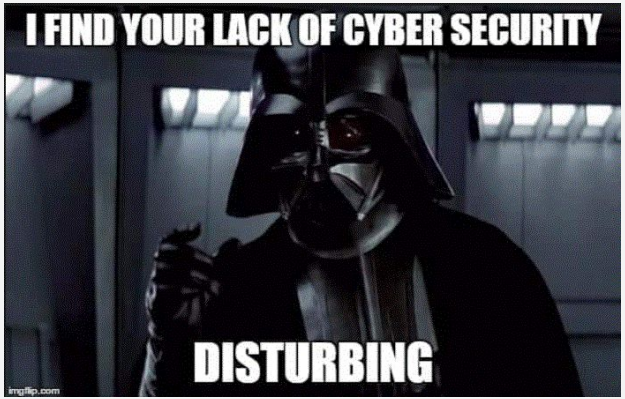
$25 \leq \alpha \leq 30$

$0 \leq \beta \leq 25$

$\alpha + \beta - \gamma \leq 50$

Security Penalties

Stack traces visible to users:	-5 pts
Application crashes on invalid input:	-5 pts
Hardcoded credentials:	-3 pts
No input sanitization:	-3 pts
Exception propagation leaks:	-3 pts
Poor logging practices:	-2 pts
Minor issues:	-1 pt each



Evaluation Breakdown

Total: $\alpha + \beta - \gamma$

Constraints

$25 \leq \alpha \leq 30$

$0 \leq \beta \leq 25$

$\alpha + \beta - \gamma \leq 50$

Minimum grade to approve the project is 25 points.

Maximum grade obtained with at least 50 points.

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Institute of Technology